

Probabilistic Initial Orbit Determination Using Gaussian Mixture Models

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The most complete description of the state of a system at any time is given by knowledge of the probability density function, which describes the locus of possible states conditioned on any available measurement information. When employing optical data, the concept of the admissible region provides a physics-based region of the range/range-rate space that produces Earth-bound orbit solutions. This work develops a method that employs a probabilistic interpretation of the admissible region and approximates the admissible region by a Gaussian mixture to formulate an initial orbit determination solution. The Gaussian mixture representation of the probability density function is then forecast and updated with subsequent data to iteratively refine the region of uncertainty. Simulation results are presented using synthetic data over a range of orbits, in which it is shown that the new method is consistently able to initialize a probabilistic orbit solution and provide iterative refinement via follow-on tracking.

Nomenclature

a, e	=	orbit semimajor axis and eccentricity
$ A $	=	determinant of the (square) matrix A
$\ a\ $	=	two-norm of the vector a
$a \cdot b$	=	inner (dot) product between two vectors a and b
$a \times b$	=	cross product between two vectors a and b
$\mathcal{E}$	=	specific orbital energy
$f(x(t), t)$	=	nonlinear dynamical system evaluated at state $x(t)$
h	=	specific orbital angular momentum
m	=	mean of a Gaussian distribution
P	=	covariance of a Gaussian distribution
$p_g(x; m, P)$	=	Gaussian probability density function in x
$q, \dot{q}$	=	inertial position and velocity of the ground station with respect to the planet center
$r, \dot{r}$	=	inertial position and velocity of an object with respect to the planet center
R_e	=	equatorial radius of the Earth, 6.3781363×10^3 km
$x(t)$	=	state of the dynamical system at time t
α_i	=	weight of the i th component in a Gaussian mixture model
μ	=	gravitational mass parameter of the Earth, 3.986004415×10^5 km ³ /s ²
$\rho, \dot{\rho}$	=	inertial position and velocity of and object with respect to the ground station
ρ, α, δ	=	topocentric range, right ascension, and declination
$\dot{\rho}, \dot{\alpha}, \dot{\delta}$	=	time rate of change of topocentric range, right ascension, and declination
ω_e	=	angular velocity of the Earth, 7.292115×10^{-5} rad/s

I. Introduction

FOR any Bayesian estimation strategy, a method for obtaining a probability distribution of the initial state is required. One approach used in many orbit determination applications is to perform an initial orbit determination (IOD) step and then refine the initial estimate provided by IOD via a batch processor mechanism (see [1] for details on the batch processor) before transitioning to sequential estimation [e.g., an extended Kalman filter (EKF) or an unscented Kalman filter (UKF) [2,3]]. The classical methods for performing IOD are Laplace's method [4], Gauss's method [4], double r-iteration [4], or Gooding's method [5]. All of the aforementioned IOD methods require the use of measurements comprising at least six independent parameters in order to produce a six-parameter orbit.

Recently, a new class of IOD mechanisms based on the use of angle and angle-rate data has emerged. The consideration of angle-rate data in conjunction with angle data is motivated by the appearance of "photon counting" optical detectors [6]. These detectors are able to record picosecond-level time values and arc-second-level direction measurements of the arrival of individual photons on the telescope focal plane at rates of hundreds of detections per second. For most space objects (SOs) of interest, the result is an abundant data set from which high-quality derived angle and angle-rate measurements can be obtained due to the raw data density. In principle, if the motion of the observing station is known and if both the apparent direction and rate of change of direction of a satellite at two times are known, then candidate orbit solutions may be obtained via a two-body integral linkage method [7]. For the case in which the apparent direction and rate of change of direction of the satellite is known at only one time, then the position and velocity vectors of the satellite at that time are functions of only two parameters, say, range and range rate. By considering the range/range-rate space, an IOD strategy may make use of the admissible region approach of Milani et al. [8] and Tommei et al. [9], which determines the set of range/range-rate pairs such that the resultant SO orbit is Earth captured. Maruskin et al. [10] illustrated the use of the admissible region via mapping a discretization into Delaunay variables. Building upon the work of Maruskin, Fujimoto and Scheeres [11] investigated the correlation of optical observations via mapping a discretization of the admissible region into Poincaré orbit element space. DeMars et al. [12] employed a discretization of the admissible region in order to develop a multiple-hypothesis filter approach to the short-arc IOD problem. In each case, discretization of the admissible region plays an elemental role in the work.

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In this paper, the admissible region is interpreted probabilistically, and an equivalent Gaussian mixture representation is developed in order to apply a Gaussian mixture estimation algorithm to the problem of IOD and follow-on tracking in a unified and consistent framework. In doing so, the need for the “binlike” discretization of the admissible region employed in [11,12] is removed, and a fully probabilistic representation of the admissible region is obtained. The general methodology for developing the Gaussian mixture representation is based on the observation that, given the admissible region and no additional information, no single combination of range and range rate can be said to be more probable than any other combination. In probabilistic terms, this means that the admissible region can be described by a two-dimensional (2-D) uniform distribution.

The rest of the paper is organized as follows. In Sec. II, a review of the admissible region concept is given and constraints that can be applied to the admissible region are developed. Section III develops a method by which a univariate uniform distribution can be approximated by a Gaussian mixture probability density function (PDF) and applies it to the admissible region in order to formulate a probabilistic IOD solution. In Sec. IV, a Gaussian mixture formulation of the UKF is presented, including a discussion on the problem formulation, initialization, propagation of uncertainty, and the processing of measurement data. Section V then discusses the application of the proposed method to test cases involving synthetic measurement data to assess the performance of the method. Finally, Sec. VI gives some concluding remarks.

II. Review of the Admissible Region

Herein, a review of the admissible region for optical observations is presented, including the development of constraints that can be applied to the standard admissible region approach to limit the resultant size of the range/range-rate plane that must be considered. For a more detailed discussion on the admissible region, see [8,9].

It is assumed that the topocentric right ascension α , the declination δ , the time rate of change of the right ascension $\dot{\alpha}$, and the time rate of change of the declination $\dot{\delta}$ are made available via an optical observation. Consider the two-body energy, given by

$$\mathcal{E} = \frac{\|\dot{\mathbf{r}}\|^2}{2} - \frac{\mu}{\|\mathbf{r}\|} \quad (1)$$

where μ is the gravitational parameter of the central body, $\mathbf{r}$ is the inertial position of the object with respect to the planet center, and $\dot{\mathbf{r}}$ is the inertial velocity of the object with respect to the planet center. Because the optical observation is made from a ground station, the position of the object with respect to the planet center is given as the sum of the position of the ground station and the position of the object with respect to the station, and likewise for the velocities:

$$\mathbf{r} = \mathbf{q} + \boldsymbol{\rho}$$

and

$$\dot{\mathbf{r}} = \dot{\mathbf{q}} + \dot{\boldsymbol{\rho}}$$

where $\mathbf{q}$ is the inertial position of the ground station, $\dot{\mathbf{q}}$ is the inertial velocity of the ground station, $\boldsymbol{\rho}$ is the inertial position of the object with respect to the station, and $\dot{\boldsymbol{\rho}}$ is the inertial velocity of the object with respect to the station. Now, let the position and the velocity of the object with respect to the station be given in the spherical coordinates of range ρ , right ascension α , declination δ , and their time rates of change, such that

$$\boldsymbol{\rho} = \rho \mathbf{u}_\rho$$

and

$$\dot{\boldsymbol{\rho}} = \dot{\rho} \mathbf{u}_\rho + \rho \dot{\alpha} \mathbf{u}_\alpha + \rho \dot{\delta} \mathbf{u}_\delta$$

where the vectors of $\mathbf{u}_\rho$, $\mathbf{u}_\alpha$, and $\mathbf{u}_\delta$ are given by

$$\mathbf{u}_\rho = \begin{bmatrix} \cos \alpha \cos \delta \\ \sin \alpha \cos \delta \\ \sin \delta \end{bmatrix}, \quad \mathbf{u}_\alpha = \begin{bmatrix} -\sin \alpha \cos \delta \\ \cos \alpha \cos \delta \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{u}_\delta = \begin{bmatrix} -\cos \alpha \sin \delta \\ -\sin \alpha \sin \delta \\ \cos \delta \end{bmatrix}$$

Defining a set of scalar values as

$$w_0 = \|\mathbf{q}\|^2, \quad w_1 = 2(\dot{\mathbf{q}} \cdot \mathbf{u}_\rho), \quad w_2 = \dot{\alpha}^2 \cos^2 \delta + \dot{\delta}^2, \\ w_3 = 2\dot{\alpha}(\dot{\mathbf{q}} \cdot \mathbf{u}_\alpha) + 2\dot{\delta}(\dot{\mathbf{q}} \cdot \mathbf{u}_\delta), \quad w_4 = \|\dot{\mathbf{q}}\|^2, \quad \text{and} \quad w_5 = 2(\mathbf{q} \cdot \mathbf{u}_\rho)$$

the squared Euclidean norms of the position and velocity of the object with respect to the planet center are found to be

$$\|\mathbf{r}\|^2 = \rho^2 + w_5 \rho + w_0 \quad (2a)$$

$$\|\dot{\mathbf{r}}\|^2 = \dot{\rho}^2 + w_1 \dot{\rho} + w_2 \rho^2 + w_3 \rho + w_4 \quad (2b)$$

Substituting Eqs. (2) into the energy equation of Eq. (1) results in an expression for twice the energy as

$$2\mathcal{E} = \dot{\rho}^2 + w_1 \dot{\rho} + F(\rho) \quad (3)$$

where

$$F(\rho) = w_2 \rho^2 + w_3 \rho + w_4 - \frac{2\mu}{\sqrt{\rho^2 + w_5 \rho + w_0}}$$

Equation (3) may be rewritten in standard quadratic form by subtracting $2\mathcal{E}$ from both sides, giving

$$\dot{\rho}^2 + w_1 \dot{\rho} + F(\rho) - 2\mathcal{E} = 0$$

which, given a value of ρ , may be solved for $\dot{\rho}$ yielding two solutions as

$$\dot{\rho} = -\frac{w_1}{2} \pm \sqrt{\left(\frac{w_1}{2}\right)^2 - F(\rho) + 2\mathcal{E}} \quad (4)$$

A. Constraints on the Admissible Region

It is sometimes desired to add constraints to the admissible region in order to reduce the possible combinations of range/range-rate pairs that lead to permissible orbit solutions. Two constraints are described here, including constraints on the semimajor axis of the orbit and constraints on the eccentricity of the orbit:

1) The semimajor axis constraint is defined as one potential constraint that can be added to the admissible region. Equivalently, the semimajor axis constraint can be cast as an energy constraint since the two are related by

$$\mathcal{E} = -\frac{\mu}{2a} \quad (5)$$

where a is the semimajor axis. By setting a value for the semimajor axis, an equivalent energy value may be determined from Eq. (5). Then, by using this value of energy, Eq. (4) may be solved for range-rate given range, which yields a curve of constant semimajor axis in the range/range-rate space.

2) The eccentricity constraint is defined as another potential constraint. To develop the eccentricity constraint, first consider the specific angular momentum as

$$\mathbf{h} = \mathbf{r} \times \dot{\mathbf{r}}$$

Defining the vector parameters

$$\begin{aligned} \mathbf{h}_1 &= \mathbf{q} \times \mathbf{u}_\rho, & \mathbf{h}_2 &= \mathbf{u}_\rho \times (\dot{\alpha}\mathbf{u}_\alpha + \dot{\delta}\mathbf{u}_\delta), \\ \mathbf{h}_3 &= \mathbf{u}_\rho \times \dot{\mathbf{q}} + \mathbf{q} \times (\dot{\alpha}\mathbf{u}_\alpha + \dot{\delta}\mathbf{u}_\delta), & \text{and } \mathbf{h}_4 &= \mathbf{q} \times \dot{\mathbf{q}} \end{aligned}$$

it can be shown that the specific angular momentum is given by

$$\mathbf{h} = \mathbf{h}_1\dot{\rho} + \mathbf{h}_2\rho^2 + \mathbf{h}_3\rho + \mathbf{h}_4$$

Defining the scalar parameters

$$\begin{aligned} c_0 &= \|\mathbf{h}_1\|^2, & c_1 &= 2\mathbf{h}_1 \cdot \mathbf{h}_2, & c_2 &= 2\mathbf{h}_1 \cdot \mathbf{h}_3, \\ c_3 &= 2\mathbf{h}_1 \cdot \mathbf{h}_4, & c_4 &= \|\mathbf{h}_2\|^2, & c_5 &= 2\mathbf{h}_2 \cdot \mathbf{h}_3, \\ c_6 &= 2\mathbf{h}_2 \cdot \mathbf{h}_4 + \|\mathbf{h}_3\|^2, & c_7 &= 2\mathbf{h}_3 \cdot \mathbf{h}_4, & \text{and } c_8 &= \|\mathbf{h}_4\|^2 \end{aligned}$$

it can be shown that the squared Euclidean norm of the specific angular momentum is given by

$$\|\mathbf{h}\|^2 = c_0\dot{\rho}^2 + P(\rho)\dot{\rho} + U(\rho) \quad (6)$$

where

$$\begin{aligned} P(\rho) &= c_1\rho^2 + c_2\rho + c_3 \\ U(\rho) &= c_4\rho^4 + c_5\rho^3 + c_6\rho^2 + c_7\rho + c_8 \end{aligned}$$

The eccentricity is related to both the specific angular momentum and specific energy by [13]

$$e = \sqrt{1 + \frac{2\mathcal{E}\|\mathbf{h}\|^2}{\mu^2}}$$

which may be rearranged as

$$2\mathcal{E}\|\mathbf{h}\|^2 = -\mu^2(1 - e^2) \quad (7)$$

Substituting for $2\mathcal{E}$ from Eq. (3) and for $\|\mathbf{h}\|^2$ from Eq. (6) into Eq. (7), it follows that

$$(\dot{\rho}^2 + w_1\dot{\rho} + F(\rho))(c_0\dot{\rho}^2 + P(\rho)\dot{\rho} + U(\rho)) = -\mu^2(1 - e^2)$$

which may be rewritten as

$$a_4\dot{\rho}^4 + a_3\dot{\rho}^3 + a_2\dot{\rho}^2 + a_1\dot{\rho} + a_0 = 0 \quad (8)$$

where

$$\begin{aligned} a_4 &= c_0, & a_3 &= P(\rho) + c_0w_1, & a_2 &= U(\rho) + c_0F(\rho) + w_1P(\rho), \\ a_1 &= F(\rho)P(\rho) + w_1U(\rho), & \text{and } a_0 &= F(\rho)U(\rho) + \mu^2(1 - e^2) \end{aligned}$$

Therefore, given a value of eccentricity, a curve of constant eccentricity may be determined by solving for the roots of Eq. (8), which is a quartic equation in $\dot{\rho}$ for a given value of ρ . Because Eq. (8) is quartic in $\dot{\rho}$, four solutions will be obtained in solving for the roots. Any imaginary solutions that result are discarded and only the real solutions are considered when determining the curve of constant eccentricity.

B. Example of the Admissible Region and Associated Constraints

To illustrate the determination of the admissible region and the semimajor axis and eccentricity constraints, consider an optical observation of $\alpha = 10^\circ$, $\delta = -2^\circ$, $\dot{\alpha} = 15^\circ/\text{h}$, and $\dot{\delta} = 3^\circ/\text{h}$. In this example, the inertial ground station is taken to be on the surface of the Earth (i.e., $\|\mathbf{q}\| = R_e$) and to have spherical coordinates of $\varphi = 30^\circ$ (as measured from the equatorial plane) and $\lambda = 0^\circ$ (as measured from the inertial x axis), and the inertial velocity is given by $\dot{\mathbf{q}} = \boldsymbol{\omega} \times \mathbf{q}$, where $\boldsymbol{\omega} = [0 \ 0 \ \omega_e]^T$ is the angular velocity vector of the Earth. Then, the admissible region is determined by setting $\mathcal{E} = 0$ and solving Eq. (4) for $\dot{\rho}$ given values of

the range ρ . This then yields the admissible region shown in Fig. 1a. The shaded region in Fig. 1a describes the possible combinations of range and range rate that permit the object to be in an Earth captured orbit.

The determination of the constrained admissible region results by imposing a semimajor axis constraint along with an eccentricity constraint (and any other desired constraints) on the unconstrained admissible region, i.e., the region described in Fig. 1a. In this example, the semimajor axis constraint is chosen to be $a \leq 50,000$ km, and the eccentricity constraint is chosen to be $e \leq 0.4$. The preceding procedure for determining the curves of constant semimajor axis and constant eccentricity are applied, resulting in the curves shown in Fig. 1b, which are overlaid on the unconstrained admissible region. The curve of constant semimajor axis is shown in black, and the curve of constant eccentricity is shown in dark gray. The region associated with orbits that have semimajor axis less than the constraint value is the region that encompasses the (0,0) point of the range/range-rate plane and that does not go beyond the curve of constant semimajor axis. Similarly, the region associated with orbits that have eccentricity less than the constraint value is the region inside of the closed curve of constant eccentricity. By taking the intersection of these two regions, the constrained admissible region may be obtained, and this region is shown by the darker region of Fig. 1c.

III. Approximating the Admissible Region with a Gaussian Mixture

Given the (constrained) admissible region and no additional information, no single combination of range and range rate is more likely than any other, meaning that the admissible region can be interpreted probabilistically as describing a uniform distribution with the support defined by the boundaries of the (constrained) admissible region. For this reason, a method for describing a uniform distribution via a Gaussian mixture model is now developed. First, the problem of approximating a univariate uniform distribution is discussed, followed by the application of the method to generating a Gaussian mixture approximation of the admissible region.

A. Gaussian Mixture Approximation of a Univariate Uniform Distribution

The problem of approximating a univariate uniform distribution using a Gaussian mixture PDF is now examined. It is desired to approximate the uniform PDF

$$p(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

by a Gaussian mixture PDF of the form

$$q(x) = \sum_{\ell=1}^L \alpha_\ell p_g(x; m_\ell, P_\ell) \quad (10)$$

where $p_g(x; a, A)$ represents a Gaussian PDF for the variable x with mean a and covariance A ; that is,

$$p_g(x; a, A) = |2\pi A|^{-1/2} \exp\left\{-\frac{1}{2}(x-a)^T A^{-1}(x-a)\right\}$$

To facilitate the approximation, the distance between $p(x)$ and $q(x)$ is taken as the L_2 norm via

$$L_2[p\|q] = \int_{\mathbb{R}} (p(x) - q(x))^2 dx \quad (11)$$

and the optimization problem

$$\min J = L_2[p\|q] \text{ subject to } \alpha_\ell \geq 0 \ \forall \ell \text{ and } \sum_{\ell=1}^L \alpha_\ell = 1 \quad (12)$$

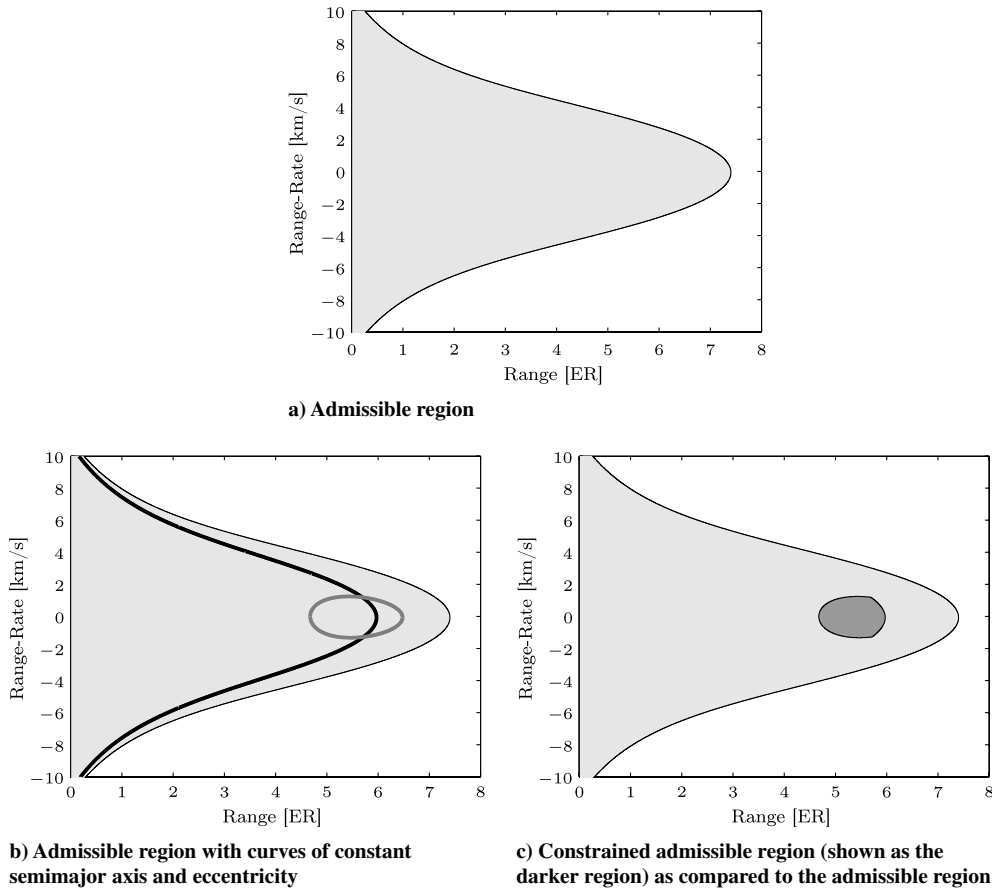


Fig. 1 Region of admissible range/range-rate values given angle and angle-rate data. Units of ER represent Earth radii.

is considered in order to determine the weights, means, and variances of the components in the Gaussian mixture approximation. It should be noted that the number of components in the Gaussian mixture is assumed to be specified so that it is not a parameter of the optimization problem.

To make the optimization problem of Eq. (12) more computationally tractable, the L_2 distance between the uniform and Gaussian mixture PDFs needs to be cast in closed form. Noting that $p(x) = 0 \forall x \in \mathbb{R} \setminus [a, b]$ Eq. (11) can be written as

$$L_2[p||q] = \int_a^b p^2(x) dx + \int_{-\infty}^{\infty} q^2(x) dx - 2 \int_a^b p(x)q(x) dx$$

Substituting for $p(x)$ from Eq. (9) then yields

$$L_2[p||q] = \frac{1}{b-a} + \int_{-\infty}^{\infty} q^2(x) dx - \frac{2}{b-a} \int_a^b q(x) dx \quad (13)$$

From the property that the multiplication of two Gaussian PDFs is given by a scaled Gaussian PDF as [14]

$$p_g(x; a, A)p_g(x; b, B) = \Gamma(a, b, A, B)p_g(x; c, C) \quad (14)$$

where

$$c = C(A^{-1}a + B^{-1}b), \quad C = (A^{-1} + B^{-1})^{-1}, \quad \text{and}$$

$$\Gamma(a, b, A, B) = |2\pi(A+B)|^{-1/2} \exp\left\{-\frac{1}{2}(a-b)^T(A+B)^{-1}(a-b)\right\}$$

it is straightforward to show that

$$\int_{-\infty}^{\infty} q^2(x) dx = \sum_{i=1}^L \sum_{j=1}^L \alpha_i \alpha_j \Gamma(m_i, m_j, P_i, P_j) \quad (15)$$

Defining the error function as [15]

$$\text{erf}\{z\} = \frac{2}{\sqrt{\pi}} \int_0^z \exp\{-t^2\} dt$$

the cumulative distribution function (CDF) for a Gaussian PDF can be expressed in terms of the error function as

$$\int_{-\infty}^z p_g\left(x; \mu, \Sigma\right) = \frac{1}{2} \left[1 + \text{erf}\left\{\frac{z-\mu}{\sqrt{2\Sigma}}\right\} \right] \quad (16)$$

By applying the CDF of a Gaussian distribution in Eq. (16) to the Gaussian mixture model (GMM) PDF of Eq. (10), it follows that

$$\int_a^b q(x) dx = \frac{1}{2} \sum_{\ell=1}^L \alpha_{\ell} \left[\text{erf}\left\{\frac{b-m_{\ell}}{\sqrt{2P_{\ell}}}\right\} - \text{erf}\left\{\frac{a-m_{\ell}}{\sqrt{2P_{\ell}}}\right\} \right] \quad (17)$$

Substituting Eqs. (15) and (17) into Eq. (13) yields

$$L_2[p||q] = \frac{1}{b-a} + \sum_{i=1}^L \sum_{j=1}^L \alpha_i \alpha_j \Gamma(m_i, m_j, P_i, P_j) - \frac{1}{b-a} \sum_{\ell=1}^L \alpha_{\ell} \left[\text{erf}\left\{\frac{b-m_{\ell}}{\sqrt{2P_{\ell}}}\right\} - \text{erf}\left\{\frac{a-m_{\ell}}{\sqrt{2P_{\ell}}}\right\} \right]$$

which, modulo the appearance of the error function, is a closed-form expression for the L_2 distance between an arbitrary uniform PDF and an arbitrary GMM PDF. With this form of the L_2 distance, the

optimization problem in Eq. (12) is ill conditioned due to a large number of local minima that can appear by exchanging two components of the GMM PDF with no change in the L_2 distance. Additionally, there are $3L$ parameters that need to be found via the optimization, which even for small L yields a larger than desired search space. To help reduce the number of parameters and create a better-conditioned optimization problem, the following are assumed:

- 1) The weights are equal for all components.
- 2) The means are evenly distributed across the support of $p(x)$.
- 3) The Gaussian mixture is homoscedastic.

Making the preceding assumptions reduces the L_2 distance to

$$L_2[p||q] = \frac{1}{b-a} + \frac{\alpha^2}{2\sqrt{\pi}\sigma} \sum_{i=1}^L \sum_{j=1}^L \exp\left\{-\frac{1}{4}\left(\frac{m_i - m_j}{\sigma}\right)^2\right\} - \frac{\alpha}{b-a} \sum_{\ell=1}^L [\operatorname{erf}\{B_\ell\} - \operatorname{erf}\{A_\ell\}]$$

where

$$A_\ell = \left(\frac{a - m_\ell}{\sqrt{2}\sigma}\right) \quad \text{and} \quad B_\ell = \left(\frac{b - m_\ell}{\sqrt{2}\sigma}\right)$$

and requires optimization over only the common standard deviation parameter, σ . Furthermore, the constraints in the optimization problem of Eq. (12), namely, that the weights all be positive and sum to one, are automatically satisfied. The optimization problem is now readily cast as a root finding problem by satisfaction of the first-order optimality conditions, i.e.,

$$\frac{dL_2[p||q]}{d\sigma} = 0 \quad (18)$$

where the derivative of the L_2 distance with respect to the standard deviation parameter is given by

$$\frac{dL_2[p||q]}{d\sigma} = \frac{\alpha^2}{2\sqrt{\pi}\sigma^2} \sum_{i=1}^L \sum_{j=1}^L \left[\frac{1}{2} \left(\frac{m_i - m_j}{\sigma} \right)^2 - 1 \right] \exp\left\{ \frac{1}{4} \left(\frac{m_i - m_j}{\sigma} \right)^2 \right\} - \frac{2\alpha}{(b-a)\sqrt{\pi}\sigma} \sum_{\ell=1}^L [A_\ell \exp\{-A_\ell^2\} - B_\ell \exp\{-B_\ell^2\}] \quad (19)$$

Then, by checking to ensure that the second derivative is positive, the candidate optimal solution found via the root finding problem of Eqs. (18) and (19) can be confirmed as the optimal solution. For brevity, the second derivative is not given, but it readily follows from Eq. (19).

The procedure for generating a library of solutions (i.e., an optimized value of σ given a number of components L) is now summarized. Without loss of generality, let the parameters of the univariate uniform PDF be $a = 0$ and $b = 1$. Specifying the number of components L and following the previously discussed rules regarding the weights and means, it follows that

$$\tilde{\alpha} = \frac{1}{L} \quad \text{and} \quad \tilde{m}_\ell = \frac{\ell}{L+1} \quad \forall \ell \in \{1, 2, \dots, L\} \quad (20)$$

The designation of $\tilde{\alpha}$ and $\tilde{m}_\ell$ is to denote that the solutions are to be obtained for the case of $a = 0$ and $b = 1$. To determine the optimal standard deviation, the root finding problem summarized by Eqs. (18) and (19) is solved, and the output solution, which satisfies Eq. (18), is denoted by $\tilde{\sigma}$. The library of solutions is then given by the pairs of L and $\tilde{\sigma}$. For instance, the values of $\tilde{\sigma}$ for $L = 1$ to $L = 15$ obtained via this procedure are summarized in Table 1. By specifying L , Eqs. (20) are used to compute the weight and mean parameters, and the library of solutions is used to determine the optimal value of $\tilde{\sigma}$. Given the values of the GMM (i.e., $\tilde{\alpha}$, $\tilde{m}_\ell$, and $\tilde{\sigma}$), which approximate the uniform distribution with $a = 0$ and $b = 1$, an arbitrary uniform distribution approximation can then be obtained via

Table 1 Example solutions for $\tilde{\sigma}$ up to $L = 15$

L	$\tilde{\sigma}$
1	0.3467
2	0.2903
3	0.2466
4	0.2001
5	0.1531
6	0.1225
7	0.1026
8	0.0884
9	0.0778
10	0.0696
11	0.0629
12	0.0575
13	0.0529
14	0.0490
15	0.0456

$$\alpha = \tilde{\alpha}, \quad m_\ell = a + (b-a)\tilde{m}_\ell, \quad \text{and} \quad \sigma = (b-a)\tilde{\sigma}$$

Using the previously discussed approach to determine the standard deviation parameter for 1-, 10-, 100-, and 1000-component Gaussian mixture PDFs so as to minimize the L_2 distance between the Gaussian mixture PDF and a uniform PDF yields the results as shown in Fig. 2, where the uniform PDF is given by the solid line and the GMM PDF is given by the dashed line. From Fig. 2, it is observed that, by increasing the number of components available in the Gaussian mixture, closer agreement to a uniform distribution is achieved.

B. Application to the Admissible Region

Because the admissible region represents a uniform PDF in two dimensions, the problem of approximating the admissible region with a Gaussian mixture must be addressed by breaking the 2-D approximation into two one-dimensional approximations so that the developed Gaussian mixture approximation strategy may be applied.

The first of the two approximations deals with the range coordinate of the admissible region. The first step is to determine the range-marginal PDF that results from interpreting the admissible region as a uniform distribution in two dimensions. That is, if $p_{\rho,\dot{\rho}}(\rho, \dot{\rho})$ represents the total uniform PDF of the admissible region, then

$$p_\rho(\rho) = \int_{-\infty}^{\infty} p_{\rho,\dot{\rho}}(\rho, \nu) d\nu \quad (21)$$

is the range-marginal PDF. In general, the admissible region boundaries are determined numerically, and so a simple numerical procedure can be employed to determine the range-marginal PDF. The support of the range-marginal PDF, defined by the scalar values a and b with $a < b$, is readily determined by the extremal range values for which the admissible region exists. To apply a set of predetermined libraries that contain the optimized GMM parameters, a maximum standard deviation for the range direction, $\sigma_{\rho,\max}$, is specified. Then, the precomputed libraries are searched to find L , the number of components, such that $\tilde{\sigma}(b-a) \leq \sigma_{\rho,\max}$. For example, let $\sigma_{\rho,\max} = 2$ and $(b-a) = 30$. In this case, it is desired to find $\tilde{\sigma}$ such that $30\tilde{\sigma} \leq 2$, or $\tilde{\sigma} \leq 1/15 = 0.0667$. From Table 1, it follows that the smallest number of components that satisfies this condition is $L = 11$. Given the determination of the number of components and hence $\tilde{\sigma}$, the parameters of the GMM approximation are

$$\alpha_\ell = \frac{1}{L}, \quad m_\ell = a + \frac{(b-a)\ell}{L+1}, \quad P_\ell = ((b-a)\tilde{\sigma})^2 \quad \forall \ell \in \{1, 2, \dots, L\} \quad (22)$$

These parameters then produce a GMM approximation to a uniform distribution with the same support set as the range-marginal PDF; they do not, however, produce an approximation to the range-marginal PDF because the range-marginal PDF is not, in general, represented by a uniform distribution.

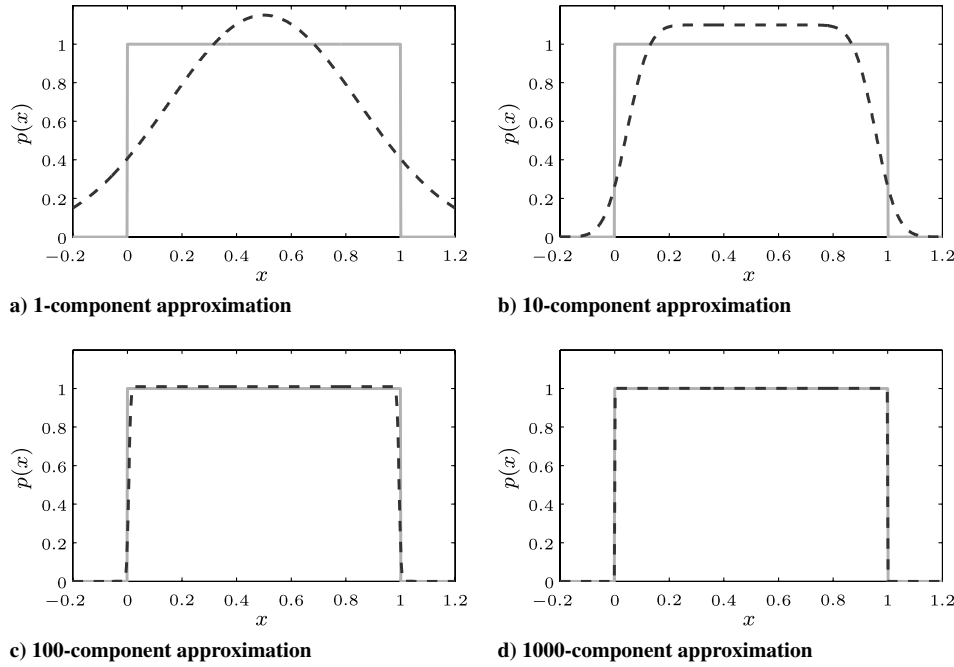


Fig. 2 Gaussian mixture approximation of a uniform PDF for different numbers of components.

To account for the fact that the range-marginal PDF is not a uniform distribution, first note that the GMM PDF of Eq. (10) is linear in the weight parameters such that, if the weights are concatenated into a vector α , the L -component GMM PDF may be expressed as

$$q(x) = \mathbf{h}^T(x)\alpha$$

where $\alpha \in \mathbb{R}^L$ and $\mathbf{h}(x) \in \mathbb{R}^L$ with the ℓ th element given by

$$h_\ell(x) = p_g(x; m_\ell, P_\ell)$$

Now, let M values of the range that lay in the support of the range-marginal PDF be taken, yielding $\rho_i \forall i \in \{1, 2, \dots, M\}$. Evaluate the range-marginal PDF at each value of the range, and set $p_i = p_\rho(\rho_i)$. Collect the evaluations of the range-marginal PDF in a vector $\mathbf{p}$, in which the i th element is p_i . Then, the weights of GMM PDF are found by solving the following least-squares problem subject to linear equality/inequality constraints:

$$\min J = \|\mathbf{p} - \mathbf{H}\alpha\| \text{ subject to } \alpha \geq \mathbf{0} \text{ and } \mathbf{1}^T \alpha = 1 \quad (23)$$

where $\mathbf{1} \in \mathbb{R}^L$ is a vector of ones, and $\mathbf{H} \in \mathbb{R}^{M \times L}$ with the element in the i th row and ℓ th column given by

$$H_{i,\ell} = p_g(\rho_i; m_\ell, P_\ell)$$

Obtaining the solution to the least-squares problem of Eq. (23) yields a set of weights such that when combined with the means and covariances of Eq. (22) produce a GMM approximation to the range-marginal PDF.

The process of obtaining a GMM approximation to the range-marginal PDF is illustrated in Fig. 3. Figure 3a shows the actual range-marginal PDF as computed from Eq. (21). Figure 3b shows the application of the GMM approximation to a uniform distribution with the same support set as the range-marginal PDF as found via Eq. (22). Figure 3c then illustrates the computation of the weight parameters via solution of the least-squares problem of Eq. (23), and Fig. 3d shows the final result of the GMM approximation to the range-marginal PDF as compared to the actual range-marginal PDF.

Up to this point, a GMM approximation for the range-marginal PDF has been obtained, but no consideration of the range-rate direction has been given. Before proceeding, consider a single

component of the range-marginal GMM approximation. The range-rate ambiguity associated with this component is uniform, meaning that given a value of range no associated value of range rate within the constrained admissible region is more or less likely. Therefore, the idea behind extending the range-marginal GMM PDF to the constrained admissible region is to apply the previously discussed GMM approximation of the uniform PDF to the range-rate direction for each component of the range-marginal GMM approximation in order to develop a bivariate GMM PDF that describes the constrained admissible region.

The first step to including the range-rate direction is to relabel the GMM approximation of the range-marginal PDF. As such, let the number of components, means, and covariances obtained in Eq. (22) be given by L_ρ , $m_{\rho,\ell}$, and $P_{\rho,\ell}$, respectively. Similarly, let the weights that were found via the least-squares problem of Eq. (23) be given by $\alpha_{\rho,\ell}$. When put together, these values form a GMM approximation of the range-marginal PDF of Eq. (21), which is given by

$$p_\rho(\rho) \approx \sum_{\ell=1}^{L_\rho} \alpha_{\rho,\ell} p_g(\rho; m_{\rho,\ell}, P_{\rho,\ell})$$

The next step is to determine the total number of components that will be in the bivariate GMM representing the constrained admissible region. Recalling that $p_{\rho,\dot{\rho}}(\rho, \dot{\rho})$ represents the total uniform PDF of the constrained admissible region, the extremal range-rate values are computed at each range value that is dictated by the means of the range-marginal GMM; that is, for each $m_{\rho,\ell}$, compute

$$a_\ell = \arg \min_{\dot{\rho}} p_{\rho,\dot{\rho}}(m_{\rho,\ell}, \dot{\rho})$$

and

$$b_\ell = \arg \max_{\dot{\rho}} p_{\rho,\dot{\rho}}(m_{\rho,\ell}, \dot{\rho})$$

which are used to define $\Delta_\rho = (b_\ell - a_\ell)$. To apply the predetermined libraries (refer to Table 1 for an example) of optimized GMM parameters that approximate a uniform PDF, a maximum standard deviation for the range-rate direction, $\sigma_{\dot{\rho},\max}$, is specified. Then, the precomputed libraries are searched to find $L_{\dot{\rho},\ell}$, the number of components, such that $\tilde{\sigma}\Delta_\rho \leq \sigma_{\dot{\rho},\max}$. Then, the total number of components of the bivariate GMM approximation of the constrained admissible region is given by $L = \sum_{\ell=1}^{L_\rho} L_{\dot{\rho},\ell}$.

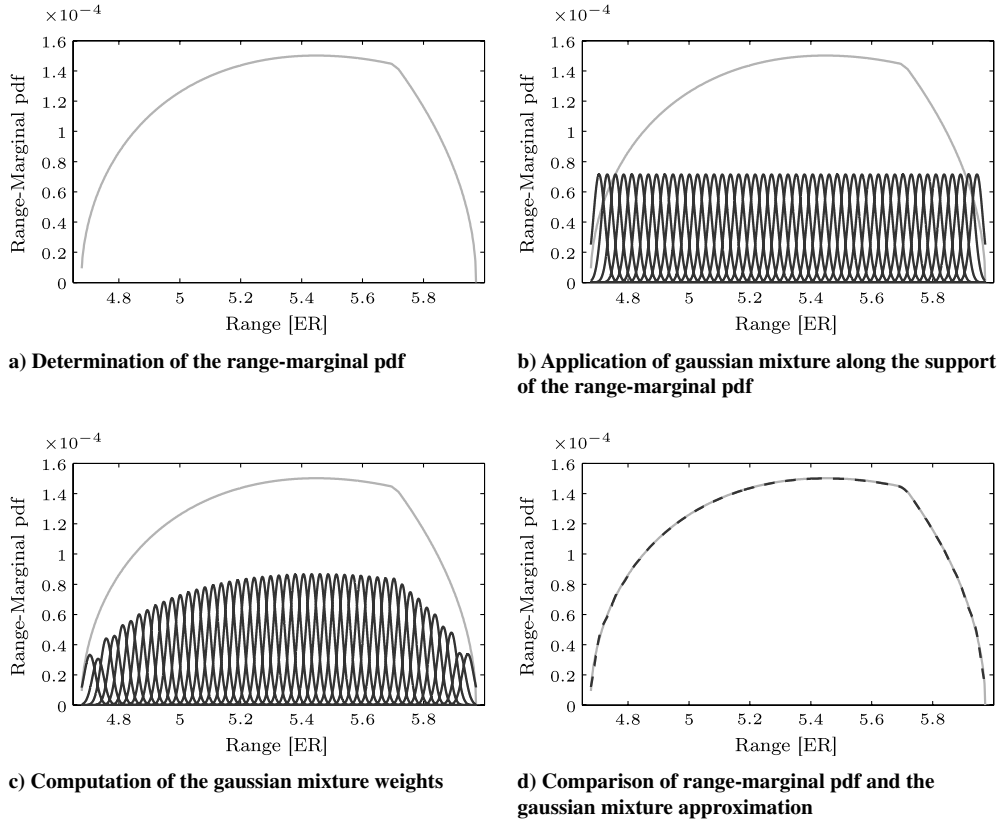


Fig. 3 Illustration of the process for computing a Gaussian mixture approximation of the range-marginal PDF.

The final step is to compute the parameters of the L -component bivariate GMM derived from the GMM parameters of the range-marginal PDF and application of the GMM approximation of a uniform PDF to the range-rate direction. Recall that, for each component of the range-marginal GMM PDF, the range-rate direction ambiguity is uniform. Therefore, for the ℓ th component of the range-marginal PDF, the range-rate uniform PDF is approximated by an $L_{\dot{\rho},\ell}$ -component GMM with parameters

$$\alpha_{\dot{\rho},k} = \frac{1}{L_{\dot{\rho},\ell}}, \quad m_{\dot{\rho},k} = a_{\ell} + \frac{\Delta_{\ell}k}{L_{\dot{\rho},\ell} + 1}, \quad \text{and} \\ P_{\dot{\rho},k} = (\Delta_{\ell}\tilde{\sigma})^2 \quad \forall k \in \{1, 2, \dots, L_{\dot{\rho},\ell}\} \quad (24)$$

Let the indexing variable i_{ℓ} be defined as $i_{\ell} = \sum_{j=1}^{\ell-1} L_{\dot{\rho},j}$. Then, given the parameters of the ℓ th component of the range-marginal GMM approximation and the k -associated components of the range-rate GMM approximation, the corresponding components of the bivariate GMM approximation of the constrained admissible region are

$$\alpha_{\rho,\dot{\rho},i_{\ell}+k} = \alpha_{\rho,\ell}\alpha_{\dot{\rho},k}, \quad \mathbf{m}_{\rho,\dot{\rho},i_{\ell}+k} = \begin{bmatrix} m_{\rho,\ell} \\ m_{\dot{\rho},k} \end{bmatrix}, \\ \text{and } \mathbf{P}_{\rho,\dot{\rho},i_{\ell}+k} = \begin{bmatrix} P_{\rho,\ell} & 0 \\ 0 & P_{\dot{\rho},k} \end{bmatrix} \quad (25)$$

where $k \in \{1, 2, \dots, L_{\dot{\rho},\ell}\}$. The preceding procedure is then repeated for every component of the range-marginal PDF such that the end result is a GMM approximation of the constrained admissible region of the form

$$p_{\rho,\dot{\rho}}(\rho, \dot{\rho}) \approx \sum_{i=1}^L \alpha_{\rho,\dot{\rho},i} p_g(\rho, \dot{\rho}; \mathbf{m}_{\rho,\dot{\rho},i}, \mathbf{P}_{\rho,\dot{\rho},i}) \quad (26)$$

Applying the entire procedure for generating a Gaussian mixture approximation of the admissible region to the constrained admissible region of Fig. 1c yields the PDF approximation illustrated in Fig. 4b, which has components located at the indicated points in Fig. 4a.

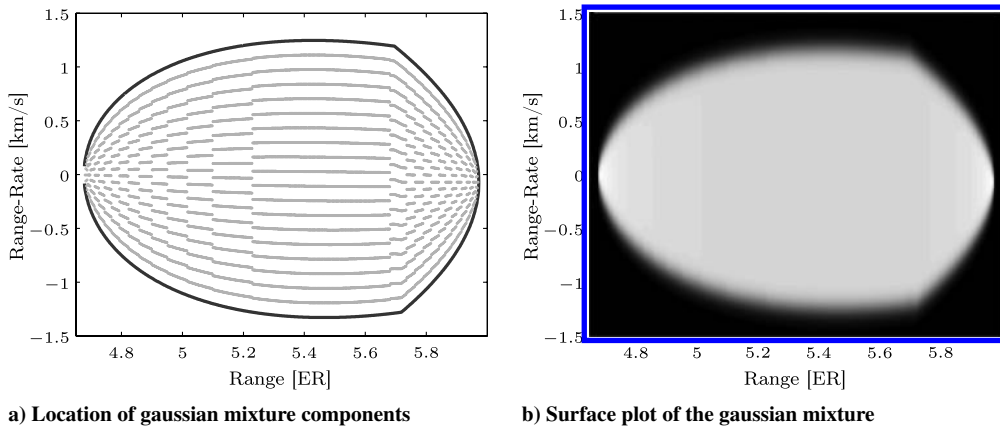


Fig. 4 Gaussian mixture representation of the constrained admissible region.

Comparing Fig. 4b to Fig. 1c, it is seen that the Gaussian mixture approximation represents the overall shape of the constrained admissible region and yields a constant value of the PDF.

IV. Gaussian Mixture Unscented Kalman Filter

A. Problem Formulation

Consider the nonlinear dynamical system governed by the differential equation

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), t), \quad \mathbf{x}(t_0) = \mathbf{x}_0 \quad (27)$$

where $\mathbf{x}(t)$ is the state of the system, $\mathbf{f}(\cdot)$ represents the nonlinear dynamics of the system, and $\mathbf{x}_0$ is the initial condition, which is assumed to be random with PDF $p(\mathbf{x}_0)$. The lack of a process noise term in Eq. (27) is justified on the basis of the nature of the problem considered in this work, wherein any noise that may be present is insignificant in comparison to the natural dynamics of the problem. The state of the system is indirectly observed by discrete-time nonlinear imperfect measurements at time t_k , which are described by

$$\mathbf{y}_k = \mathbf{h}(\mathbf{x}_k, t_k) + \mathbf{v}_k \quad (28)$$

where $\mathbf{x}_k$ is the state of the system at time t_k , $\mathbf{h}(\cdot)$ represents the nonlinear measurement function, and $\mathbf{v}_k$ is the measurement noise. The measurement noise is assumed to be a zero-mean white-noise sequence with first and second moments given, respectively, by

$$\mathbf{E}\{\mathbf{v}_k\} = \mathbf{0} \quad \text{and} \quad \mathbf{E}\{\mathbf{v}_k \mathbf{v}_{k'}^T\} = \mathbf{R}_k \delta_{kk'}$$

where $\delta_{kk'} = 0$ for $k \neq k'$ and $\delta_{kk'} = 1$ for $k = k'$.

B. Initialization

Unlike the EKF or UKF, which employ only the first two moments of the state distribution (i.e., mean and covariance), the Gaussian mixture unscented Kalman filter (GMUKF) uses a description of the PDF that is not necessarily Gaussian; instead, the GMUKF employs a GMM of the initial PDF given by

$$p(\mathbf{x}_0) = \sum_{i=1}^L \alpha_{i,0} p_g(\mathbf{x}_0; \mathbf{m}_{i,0}, \mathbf{P}_{i,0}) \quad (29)$$

where the individual Gaussian component weights are constrained such that

$$0 \leq \alpha_i \leq 1 \quad \forall i \in \{1, \dots, L\} \quad \text{and} \quad \sum_{i=1}^L \alpha_i = 1$$

The parameters of the GMM, i.e., $\mathbf{m}_i$, α_i , and $\mathbf{P}_i$, are interpreted as follows: the mean values represent the regions in the state-space in which the majority of the probability mass is concentrated; the component weights represent the normalized probability of the localized probability mass; and the covariance matrices are used to limit the regions of the state space about the means such that each term in the GMM is effectively zero outside of a neighborhood of the mean.

C. Propagation

Consider the time-propagation of the PDF. Given the time interval $t \in [t_{k-1}, t_k]$, it is desired to approximate the conditional PDF at time t_k via

$$p(\mathbf{x}_k | \mathbf{Y}^{k-1}) = \sum_{\ell=1}^L \alpha_{\ell,k}^- p_g(\mathbf{x}_k; \mathbf{m}_{\ell,k}^-, \mathbf{P}_{\ell,k}^-) \quad (30)$$

based on the starting condition at time t_{k-1}

$$p(\mathbf{x}_{k-1} | \mathbf{Y}^{k-1}) = \sum_{\ell=1}^L \alpha_{\ell,k-1}^+ p_g(\mathbf{x}_{k-1}; \mathbf{m}_{\ell,k-1}^+, \mathbf{P}_{\ell,k-1}^+)$$

Recalling that the covariance matrices of the components are used to limit the region of the state space about which each component is valid, the dynamical system local to each component may be approximated via the unscented transform (UT), thereby allowing the implementation of a UKF propagation scheme for each component while holding the component weights equal across the time step. To apply the UT to the forward propagation of each component mean and covariance, first determine the square-root factor of the component covariance matrix; that is, find $\mathbf{S}_{\ell,k-1}$ such that $\mathbf{P}_{\ell,k-1}^+ = \mathbf{S}_{\ell,k-1} \mathbf{S}_{\ell,k-1}^T$, which can be readily accomplished via a Cholesky factorization. Once the square-root factor is determined, the columns of the square-root factor, given by

$$\mathbf{S}_{\ell,k-1} = [\mathbf{s}_{\ell,1,k-1} \dots \mathbf{s}_{\ell,n,k-1}]$$

are used to determine the set of $K = 2n$ sigma points that make up the symmetric sigma-point set,[‡] such that for $i \in \{1, \dots, n\}$, each component sigma point is given by [2]

$$\begin{aligned} \mathcal{X}_{\ell,i,k-1} &= \mathbf{m}_{\ell,k-1}^+ + \sqrt{n} \mathbf{s}_{\ell,i,k-1} \\ \mathcal{X}_{\ell,i+n,k-1} &= \mathbf{m}_{\ell,k-1}^+ - \sqrt{n} \mathbf{s}_{\ell,i,k-1} \end{aligned}$$

Associated with each sigma point is a corresponding weight w_i . For the symmetric sigma-point set, the weights are given by $w_i = 1/2n$ for all sigma points. Each sigma point is then numerically integrated through the nonlinear dynamics for $t \in [t_{k-1}, t_k]$ with an initial condition of $\mathcal{X}_{\ell,i}(t_{k-1}) = \mathcal{X}_{\ell,i,k-1}$; that is,

$$\dot{\mathcal{X}}_{\ell,i}(t) = \mathbf{f}(\mathcal{X}_{\ell,i}(t), t), \quad \mathcal{X}_{\ell,i}(t_{k-1}) = \mathcal{X}_{\ell,i,k-1}$$

Additionally, to each component is the associated weight α_{ℓ} , which is held constant across each time step, or for $t \in [t_{k-1}, t_k]$, we have

$$\dot{\alpha}_{\ell}(t) = 0, \quad \alpha_{\ell}(t_{k-1}) = \alpha_{\ell,k-1}^+$$

with a final condition of $\alpha_{\ell,k}^- = \alpha_{\ell}(t_k)$. The final condition on the numerical integration of the sigma points is then given for each sigma point by $\mathcal{X}_{\ell,i,k} = \mathcal{X}_{\ell,i}(t_k)$, which can then be used to approximate the nonlinear transformation of the component means and covariances using

$$\begin{aligned} \mathbf{m}_{\ell,k}^- &= \sum_{i=1}^K w_i \mathcal{X}_{\ell,i,k} \\ \mathbf{P}_{\ell,k}^- &= \sum_{i=1}^K w_i (\mathcal{X}_{\ell,i,k} - \mathbf{m}_{\ell,k}^-)(\mathcal{X}_{\ell,i,k} - \mathbf{m}_{\ell,k}^-)^T \end{aligned}$$

Combining the component means and covariance with the final condition on the numerical integration of the component weights, the a priori GMM PDF can be evaluated with Eq. (30).

D. Update

The a posteriori PDF is found by considering the composition of the a priori PDF and the measurement PDF at time t_k , and then normalizing the result, yielding Bayes' rule as [17]

$$p(\mathbf{x}_k | \mathbf{Y}^k) = \frac{p(\mathbf{x}_k | \mathbf{Y}^{k-1}) p(\mathbf{y}_k | \mathbf{x}_k)}{\int p(\boldsymbol{\xi}_k | \mathbf{Y}^{k-1}) p(\mathbf{y}_k | \boldsymbol{\xi}_k) d\boldsymbol{\xi}_k} \quad (31)$$

From the propagation stage, the conditional PDF of the state given the previous measurements is given by

[‡]Although the symmetric sigma-point set is used here for simplicity, it should be noted that any sigma-point set for the UKF can be employed in the same manner. In fact, the approach is amenable to consideration of other point-based methods, such as that of sparse Gauss-Hermite quadrature [16].

$$p(\mathbf{x}_k | \mathbf{Y}^{k-1}) = \sum_{\ell=1}^L \alpha_{\ell,k}^- p_g(\mathbf{x}_k; \mathbf{m}_{\ell,k}^-, \mathbf{P}_{\ell,k}^-) \quad (32)$$

where the mean and covariance for each component are fully captured by the propagated sigma points, $\mathcal{X}_{\ell,i,k}$. Furthermore, the conditional PDF of the current measurement given the state at time t_k is assumed to be Gaussian⁸ such that

$$p(\mathbf{y}_k | \mathbf{x}_k) = p_g(\mathbf{y}_k; \mathbf{h}(\mathbf{x}_k, t_k), \mathbf{R}_k) \quad (33)$$

Considering the numerator of Eq. (31) and substituting for $p(\mathbf{x}_k | \mathbf{Y}^{k-1})$ from Eq. (32) and for $p(\mathbf{y}_k | \mathbf{x}_k)$ from Eq. (33) yields

$$p(\mathbf{x}_k | \mathbf{Y}^{k-1}) p(\mathbf{y}_k | \mathbf{x}_k) = \sum_{\ell=1}^L \alpha_{\ell,k}^- p_g(\mathbf{x}_k; \mathbf{m}_{\ell,k}^-, \mathbf{P}_{\ell,k}^-) p_g(\mathbf{y}_k; \mathbf{h}(\mathbf{x}_k, t_k), \mathbf{R}_k)$$

Therefore, in the same manner as Alspach and Sorenson [18], the final form of the numerator of Eq. (31) is

$$p(\mathbf{x}_k | \mathbf{Y}^{k-1}) p(\mathbf{y}_k | \mathbf{x}_k) = \sum_{\ell=1}^L \beta_{\ell,k} \alpha_{\ell,k}^- p_g(\mathbf{x}_k; \mathbf{m}_{\ell,k}^+, \mathbf{P}_{\ell,k}^+) \quad (34)$$

where

$$\beta_{\ell,k} = p_g(\mathbf{y}_k; \hat{\mathbf{y}}_{\ell,k}^-, \mathbf{P}_{\ell,y}) \quad (35a)$$

$$\mathbf{m}_{\ell,k}^+ = \mathbf{m}_{\ell,k}^- + \mathbf{K}_{\ell,k}(\mathbf{y}_k - \hat{\mathbf{y}}_{\ell,k}^-) \quad (35b)$$

$$\mathbf{P}_{\ell,k}^+ = \mathbf{P}_{\ell,k}^- - \mathbf{K}_{\ell,k} \mathbf{P}_{\ell,y} \mathbf{K}_{\ell,k}^T \quad (35c)$$

$$\mathbf{K}_{\ell,k} = \mathbf{P}_{\ell,xy} \mathbf{P}_{\ell,y}^{-1} \quad (35d)$$

To compute the expected value of $\mathbf{y}_k$ as well as its covariance and the cross covariance between the a priori state and the measurement, the UT is used. In doing so, the first step is to compute the measurement-transformed sigma points for each component as

$$\mathcal{Y}_{\ell,i,k} = \mathbf{h}(\mathcal{X}_{\ell,i,k}, t_k) \quad \forall i \in \{1, \dots, K\}$$

Then, for each component, $\hat{\mathbf{y}}_{\ell,k}^-$, $\mathbf{P}_{\ell,y}$, and $\mathbf{P}_{\ell,xy}$ are found to be

$$\hat{\mathbf{y}}_{\ell,k}^- = \sum_{i=1}^K w_i \mathcal{Y}_{\ell,i,k}$$

$$\mathbf{P}_{\ell,y} = \sum_{i=1}^K w_i (\mathcal{Y}_{\ell,i,k} - \hat{\mathbf{y}}_{\ell,k}^-)(\mathcal{Y}_{\ell,i,k} - \hat{\mathbf{y}}_{\ell,k}^-)^T + \mathbf{R}_k$$

$$\mathbf{P}_{\ell,xy} = \sum_{i=1}^K w_i (\mathcal{X}_{\ell,i,k} - \mathbf{m}_{\ell,k}^-)(\mathcal{Y}_{\ell,i,k} - \hat{\mathbf{y}}_{\ell,k}^-)^T$$

The component means and covariances can then be updated via Eq. (35). To determine the update rule for the component weights, the development of Eq. (31) must be completed. To do this, consider the denominator of Eq. (31), which is given by the integral of Eq. (34), yielding

$$\int p(\mathbf{x}_k | \mathbf{Y}^{k-1}) p(\mathbf{y}_k | \mathbf{x}_k) d\mathbf{x}_k = \sum_{\ell=1}^L \beta_{\ell,k} \alpha_{\ell,k}^- \quad (36)$$

⁸The assumption that $p(\mathbf{y}_k | \mathbf{x}_k)$ is Gaussian is merely for compactness. It is straightforward to extend the algorithm to the case in which $p(\mathbf{y}_k | \mathbf{x}_k)$ is represented by a GMM.

Combining the results of Eqs. (34) and (36) yields $p(\mathbf{x}_k | \mathbf{Y}^k)$ as

$$p(\mathbf{x}_k | \mathbf{Y}^k) = \frac{1}{\sum_{\ell=1}^L \beta_{\ell,k} \alpha_{\ell,k}^-} \sum_{\ell=1}^L \beta_{\ell,k} \alpha_{\ell,k}^- p_g(\mathbf{x}_k; \mathbf{m}_{\ell,k}^+, \mathbf{P}_{\ell,k}^+)$$

which can be rearranged to give

$$p(\mathbf{x}_k | \mathbf{Y}^k) = \sum_{\ell=1}^L \alpha_{\ell,k}^+ p_g(\mathbf{x}_k; \mathbf{m}_{\ell,k}^+, \mathbf{P}_{\ell,k}^+)$$

where the a posteriori component weights are now given by

$$\alpha_{\ell,k}^+ = \frac{\beta_{\ell,k} \alpha_{\ell,k}^-}{\sum_{j=1}^L \beta_{j,k} \alpha_{j,k}^-} \quad (37)$$

In summary, when presented with measurement data at time t_k , the GMM component weights are updated using Eq. (37), the component means are updated using Eq. (35b), and the component covariances are updated using Eq. (35c).

V. Discussion

A. Dynamical System Model

Recall from the discussion of the admissible region that $\mathbf{r}$ is taken to be the inertial position of the object with respect to the planet center, that $\mathbf{q}$ is taken to be the inertial position of the ground station, and that $\boldsymbol{\rho}$ is taken to be the inertial position of the object with respect to the ground station. For the purposes of this work, it is assumed that the object motion is solely influenced by the gravity of the Earth, which is modeled as a point-mass gravitational acceleration, that the ground station is fixed to the Earth, and that the Earth has a constant angular velocity. Under these assumptions, it follows that the acceleration of the object and of the ground station are given, respectively, by

$$\ddot{\mathbf{r}} = -\frac{\mu}{\|\mathbf{r}\|^3} \mathbf{r} \quad (38a)$$

and

$$\ddot{\mathbf{q}} = \boldsymbol{\omega} \times \boldsymbol{\omega} \times \mathbf{q} \quad (38b)$$

where μ is the gravitational parameter of the Earth and $\boldsymbol{\omega}$ is the angular velocity of the Earth as expressed in the inertial frame. Let the inertial acceleration of the object with respect to the ground station be represented by $\mathbf{a}$. From Eqs. (38), it follows that

$$\mathbf{a} = -\frac{\mu}{\|\mathbf{r}\|^3} \mathbf{r} - \boldsymbol{\omega} \times \boldsymbol{\omega} \times \mathbf{q} \quad (39)$$

Now, let x , y , and z represent the Cartesian coordinates of $\boldsymbol{\rho}$. Then, the range, right ascension, and declination are given in terms of the Cartesian coordinates as

$$\rho = \sqrt{x^2 + y^2 + z^2} \quad (40a)$$

$$\alpha = \tan^{-1} \frac{y}{x} \quad (40b)$$

$$\delta = \tan^{-1} \frac{z}{w} \quad (40c)$$

where $w^2 = x^2 + y^2$. Differentiating Eqs. (40) twice with respect to time, it can be shown that

$$\ddot{\boldsymbol{\rho}} = \rho(\ddot{\alpha}^2 \cos^2 \delta + \dot{\delta}^2) + \ddot{x} \cos \alpha \cos \delta + \ddot{y} \sin \alpha \cos \delta + \ddot{z} \sin \delta \quad (41a)$$

$$\ddot{\alpha} = \frac{1}{\rho \cos^2 \delta} [2\dot{\alpha} \cos \delta (\rho \dot{\delta} \sin \delta - \dot{\rho} \cos \delta) - \ddot{x} \sin \alpha \cos \delta + \ddot{y} \cos \alpha \cos \delta] \quad (41b)$$

$$\ddot{\delta} = \frac{1}{\rho} (-2\dot{\rho} \dot{\delta} - \rho \dot{\alpha}^2 \sin \delta \cos \delta - \ddot{x} \cos \alpha \sin \delta - \ddot{y} \sin \alpha \sin \delta + \ddot{z} \cos \delta) \quad (41c)$$

where $\ddot{x}$, $\ddot{y}$, and $\ddot{z}$ are the components of $\mathbf{a}$ from Eq. (39), such that $\mathbf{a} = [\ddot{x} \ \ddot{y} \ \ddot{z}]^T$. Equations (41) represent the second-order form of the dynamical system employed in Eq. (27), with the state of the system taken to be $\mathbf{x}(t) = [\rho(t) \ \alpha(t) \ \delta(t) \ \dot{\rho}(t) \ \dot{\alpha}(t) \ \dot{\delta}(t)]^T$.

B. Measurement Model

The measurements used are the topocentric right ascension, declination, and their time rates from a station with a known location. Therefore, because the state of the system is taken to be $\mathbf{x}(t) = [\rho(t) \ \alpha(t) \ \delta(t) \ \dot{\rho}(t) \ \dot{\alpha}(t) \ \dot{\delta}(t)]^T$, as previously described, it follows that the measurement function of Eq. (28) is

$$\mathbf{h}(\mathbf{x}_k, t_k) = \mathbf{H}\mathbf{x}_k \quad \text{with} \quad \mathbf{H} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (42)$$

where $\mathbf{x}_k = \mathbf{x}(t_k)$. The noise specifications [i.e., the statistics of $\mathbf{v}_k$ in Eq. (28)] to be used on the angle and angle-rate measurements are derived from analyzing a segment of actual data. The raw data density is approximately 500 observations/s with noise on the order of 40 arc-s standard deviation. Because the data are so dense, a least-squares fit of the data can be made in order to recover an equivalent angle and angle-rate measurement. Similarly, the least-squares fit may be used to determine the noise specifications on the derived angle and angle-rate measurement. For instance, assume that a linear curve fit is applied of the form

$$\theta(t) = \theta_0 + \dot{\theta}_0(t - t_0)$$

where θ is used to represent either right ascension or declination. Then, following the analysis performed by Maruskin et al. [10] for the case of a linear curve fit instead of a parabolic curve fit, the standard deviations for θ_0 and $\dot{\theta}_0$ can be found to be

$$\sigma_{\theta_0} = \frac{1}{\sqrt{N}} \sigma_{\theta_{\text{raw}}} \quad \text{and} \quad \sigma_{\dot{\theta}_0} = \frac{2\sqrt{3}}{\Delta T \sqrt{N}} \sigma_{\theta_{\text{raw}}}$$

where $\sigma_{\theta_{\text{raw}}}$ is the standard deviation of the raw data, N is the number of data points used in the least-squares curve fit, and ΔT is the time span of the data points used in the least-square curve fit. The derived standard deviations are only approximate, however, because the analysis of Maruskin et al. [10] holds only for the case of equispaced raw data points, which is not guaranteed in general. Given that the standard deviation on the raw data is 40 arc-s, and using a 20 s ΔT with 500 observations/s, the standard deviations for the curve fit angle and angle-rate values are found to be $\sigma_{\theta_0} = 0.40$ arc-s and $\sigma_{\dot{\theta}_0} = 0.07$ arc-s/s. The 20 s time span for fitting the raw data was chosen to achieve derived statistics on the angle measurements below 0.5 arc-s and on the angle-rate measurements below 0.1 arc-s/s. With better sensor calibration, there is potential to achieve higher accuracy with the raw data, thereby reducing the time between measurements or allowing for more accurate derived measurements.

C. Track Initialization

To implement the GMUKF algorithm, an initial GMM PDF is required as described by Eq. (29). Because no filtering solution exists before the initial acquisition of data, let the first data point be given by the measurement $\mathbf{y}_0$ with an associated covariance $\mathbf{R}_0$. The

measurement $\mathbf{y}_0$ contains data on the topocentric right ascension, declination, and their time rates, as previously discussed. Using these data, the constrained admissible region is constructed and approximated with a GMM PDF, given by Eq. (26), which represents a uniform distribution across the support of the constrained admissible region. The initial state PDF $p(\mathbf{x}_0)$ is then constructed from the admissible region PDF and the initial measurement data as

$$p(\mathbf{x}_0) = \sum_{i=1}^L \alpha_{i,0} p_g(\mathbf{x}_0; \mathbf{m}_{i,0}, \mathbf{P}_{i,0}) \quad (43)$$

where

$$\alpha_{i,0} = \alpha_{\rho,\dot{\rho},i}, \quad \mathbf{m}_{i,0} = \mathbf{W} \begin{bmatrix} \mathbf{y}_0 \\ \mathbf{m}_{\rho,\dot{\rho},i} \end{bmatrix},$$

$$\text{and} \quad \mathbf{P}_{i,0} = \mathbf{W} \begin{bmatrix} \mathbf{R}_0 & \mathbf{0} \\ \mathbf{0} & \mathbf{P}_{\rho,\dot{\rho},i} \end{bmatrix} \mathbf{W}^T \quad (44)$$

The matrix $\mathbf{W}$, which appears in Eq. (44), is a permutation matrix that maps $[\alpha \ \delta \ \dot{\alpha} \ \dot{\delta} \ \rho \ \dot{\rho}]^T$ into $[\rho \ \alpha \ \delta \ \dot{\rho} \ \dot{\alpha} \ \dot{\delta}]^T$ and is given by

$$\mathbf{W} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

D. Coordinate System Selection

The coordinate system employed in this work (range, right ascension, and declination) is motivated by the fact that measurements of the right ascension, declination, and their time rates are linear with respect to the coordinate system and by the fact that the track initialization generates an initial GMM PDF that is naturally represented in this system. One can easily consider other coordinate systems (e.g., Cartesian coordinates or Keplerian elements) by simply transforming the initial GMM PDF in Eq. (43) into a different system via linearization or the UT applied to each component of the Gaussian mixture.

E. Results on Synthetic Data

To test the proposed method for the initial orbit determination and subsequent follow-on tracking, two test cases are considered, including 1) a single orbit case with a Monte Carlo sampling of the measurement data and 2) a grid of multiple orbits with a single set of data per orbit.

For both test cases, topocentric right ascension, declination, and their time rates are simulated for five nights, with one 5 min track of data per night, and the data taken at 20 s intervals. The true measurements are subjected to zero-mean white noise with a standard deviation of 0.40 arc-s for each angle measurement and 0.07 arc-s/s for each angle-rate measurement. Also, for both test cases, the first measurement is used to generate an initial GMM PDF, which constitutes the IOD step. Then, the GMUKF estimation scheme that has been discussed is employed for the forward prediction of the PDF and assimilation of the remaining data.

For the first test case, the true orbit is characterized by

$$a = 43,000 \text{ km}, \quad e = 0.03, \quad i = 3 \text{ deg}, \quad \Omega = 0 \text{ deg},$$

$$\omega = 0 \text{ deg}, \quad M = 0 \text{ deg}$$

This orbit is then used to generate 100 sets of measurement data (i.e., the realization of the measurement noise is different for each set of data), and the initial orbit determination algorithm is applied to each set of the measurement data. Recall that the track initialization

method employs the constrained admissible region for generating the initial GMM PDF. In computing the constrained admissible region, the orbit regime under consideration is constrained to be near geosynchronous, such that the maximum semimajor axis and the maximum eccentricity are taken as $a_{\max} = 43,000$ km and $e_{\max} = 0.1$. Additionally, when computing the GMM approximation for the constrained admissible region, the tolerances of $\sigma_{\rho, \max}$ and $\sigma_{\dot{\rho}, \max}$ are used to control the resolution (i.e., the component size) of the GMM approximation. For this test case, these values are taken to be $\sigma_{\rho, \max} = 5$ km and $\sigma_{\dot{\rho}, \max} = 100$ m/s.

To assess the quality of each solution, 100,000 samples are drawn from the a priori and a posteriori GMM PDFs that are obtained via application of the track initialization and follow-on tracking (via the GMUKF scheme) and are converted to Cartesian coordinates. The sample covariance is then computed, from which the position root sum square (RSS) and velocity RSS are extracted. The RSS values are plotted for each set of the measurement data in Fig. 5, and the average RSS values over all 100 runs are plotted in Fig. 6. The vertical spikes that appear in Figs. 5 and 6 indicate the increase in uncertainty that naturally occurs between arcs of data; that is, because only 5 min of data per night is used, there is approximately a 24 h period in which the uncertainty grows. However, as more arcs of data are processed, the amount of observed increase in the uncertainty diminishes. Additionally, for all 100 sets of data, although there is some variation in the computed RSS value, all cases retain the same characteristics and exhibit a good filtering solution.

For the second test case, the orbit parameters

$$i = 3 \text{ deg}, \quad \Omega = 0 \text{ deg}, \quad \omega = 0 \text{ deg}, \quad M = 0 \text{ deg}$$

are fixed, and a grid of orbits is generated with $a \in [39,000 \text{ km}, 45,000 \text{ km}]$ and $e \in [0, 0.1]$. Then, 1000 true orbits are generated by taking a grid of 50 semimajor axis values and 20 eccentricity values. For each true orbit, a single set of measurement data is generated to which the initial orbit determination algorithm is applied. As with the first test case, the constraints on semimajor axis and eccentricity used in the generation of the admissible region are taken as $a_{\max} = 43,000$ km and $e_{\max} = 0.1$. Similarly, the tolerance controls for forming the GMM approxi-

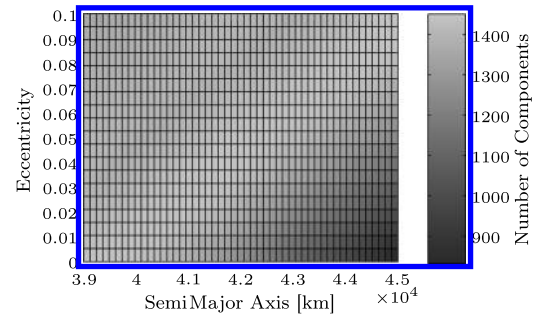
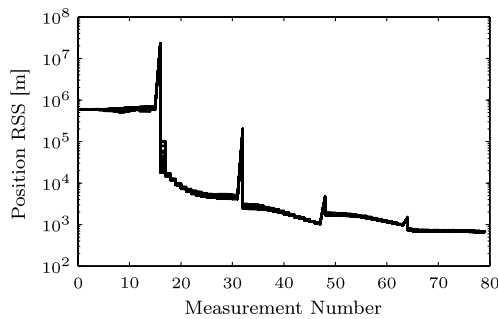


Fig. 7 Number of components in the Gaussian mixture approximation of the constrained admissible region.

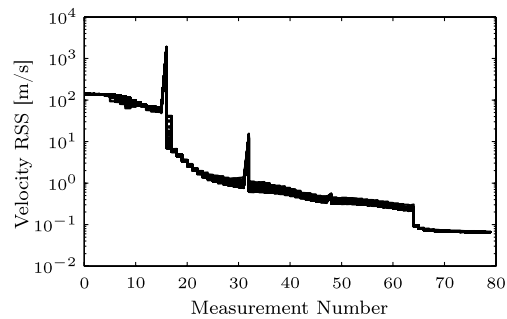
mation of the admissible region are taken to be $\sigma_{\rho, \max} = 5$ km and $\sigma_{\dot{\rho}, \max} = 100$ m/s.

The first measurement is used in conjunction with the semimajor axis and eccentricity constraints as well as the tolerance controls to formulate the constrained admissible region and its Gaussian mixture approximation. The number of components that are generated by this process varies based on the exact orbit, measurements, constraints, and tolerances applied. For the problem under consideration, the number of components obtained via the initialization process for each of the 1000 cases is shown in Fig. 7, wherein it is observed that the number of components at the initialization point ranges from 833 to 1450, with the smallest number of Gaussian mixture components occurring for the case with the highest semimajor axis and lowest eccentricity.

To assess the tracking quality on each orbit, 100,000 samples are drawn from the a posteriori GMM PDF at the time of the last data point. These are then converted to Cartesian coordinates, the sample covariance is computed, and the position and velocity RSS values are determined. The position RSS values at the final time are plotted for each of the true orbits in Fig. 8a, and the velocity RSS values at the final time are plotted for each of the true orbits in Fig. 8b. From Figs. 8a and 8b, it is observed that the maximum uncertainty over all 1000 orbits after processing the entire data set is about 1.4 km in

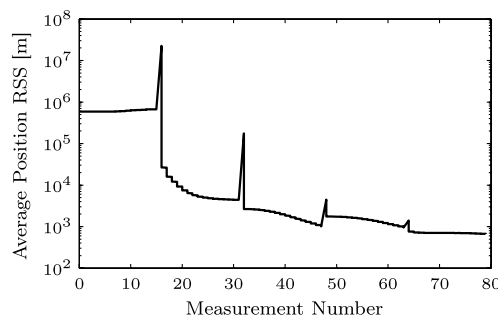


a) Position RSS

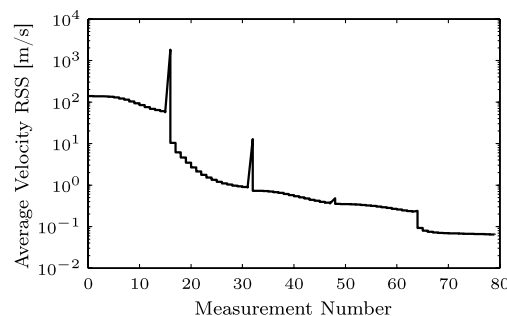


b) Velocity RSS

Fig. 5 Position and velocity RSSs as a function of measurement number for 100 runs on the same orbit.



a) Average position RSS



b) Average velocity RSS

Fig. 6 Average position and velocity RSSs as a function of measurement number for 100 runs on the same orbit.

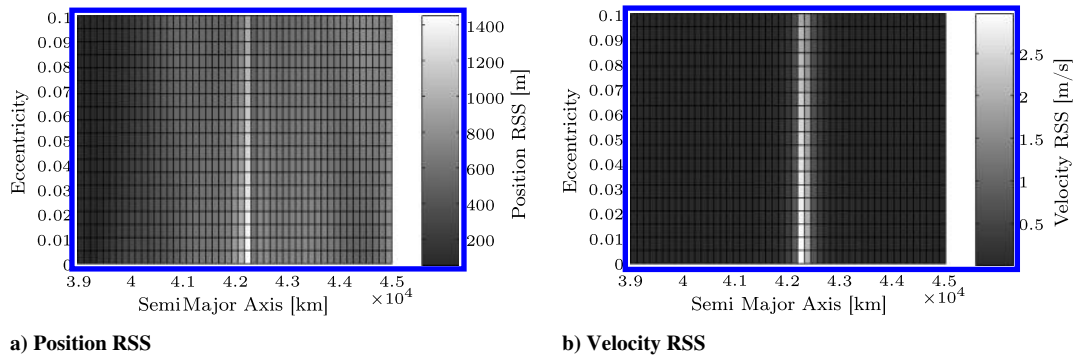


Fig. 8 Position and velocity RSSs at the final time as functions of orbit parameters.

position and about 2.7 m/s in velocity. Additionally, the minimum uncertainty is approximately 0.1 km in position and 0.2 m/s in velocity. The area in which the data produce the least amount of uncertainty resolution is confined to the nearly geosynchronous orbits, as the vertical band of higher RSS values illustrates in Figs. 8a and 8b. This artifact arises from the lack of diversity in the observational geometry, which, in turn, is a consequence of the tracking data being simulated at the same time of night for each track.

VI. Conclusions

The application of a Gaussian mixture approach to the problem of short-arc space object (SO) initial orbit determination (IOD) and follow-on tracking has been presented. In doing so, the admissible region methodology for determining the region of range/range-rate space that admits Earth-captured orbits was reviewed, and constraints on the orbit semimajor axis and eccentricity were developed. Based on the observation that, given the admissible region and no additional information, no single combination of range and range rate can be said to be more probable than any other combination, the admissible region was interpreted as a two-dimensional uniform distribution. A method for describing a uniform distribution by a Gaussian mixture was developed and applied to the admissible region in order to determine an initial Gaussian mixture model (GMM) probability density function (PDF), which comprises the IOD step. Once the initial GMM PDF is obtained, a recursive filtering strategy, i.e., the Gaussian mixture unscented Kalman filter in this work, was used to perform follow-on tracking via processing the subsequent data. The proposed method was applied to synthetic data, which showed excellent performance of the method. In particular, the method successfully tracked every set of data applied over a range of orbit parameters and over a Monte Carlo sampling of the measurement data. In scenarios in which quick follow-on tracking does not occur and the PDF can easily need to be propagated over many hours or days, discretization schemes like the multiple-hypothesis filter can easily fail, rendering methods such as the presented Gaussian mixture approach for short-arc SO IOD and follow-on tracking necessary.

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